MAHATMAGANDHIUNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS)

THIRDSEMESTEREXAMINATION

(2024ADMISIONONWARDS)

MG3DSCECT201 ProgramminginC

Duration:1.5hours **MaximumMarks:**50

**Remember(K),Understand(U),Apply(A),Analyse(An),Evaluate(E),Create(C),Skill(S), Interest (I) and Appreciation (Ap)*

PartA

MultipleChoiceQuestions

Answer **All** Questions

Eachquestioncarries1mark

1	Whichofthefollowing datatypesis usedto store asinglecharacter in C? A)int B)char C) float D)double	U	1
2	Whichofthefollowing isnot avalid operatorinC? A)+ B)* C)++ D)#	K	1
3	Whichformatspecifierisused forprintingafloatingpoint numberin C A)%s B) %f C) %d D)%c	K	1
4	Whatisthe indexof thefirstelement in anarray in C? A) 0 B) 1 C)-1 D) Undefined	K	2
5	Whatisthecorrectwaytodeclareanintegerarrayofsize5in C? A) int arr; B)int arr[5]; C) int arr(5); D)array intarr[5];	U	2
6	Whichfunction isused tocopy onestring toanother in C A) strcopy()B) copystr() C) strcpy() D)stringcopy()	K	2
7	Whatisthescopeof avariable declaredinside afunction A) global B)localtothat function C)localtotheentire program D)Static	U	2
8	Whatdoesapointerstore in C A)Addressofavariable B)Valueofavariable C)Datatype D)Sizeof a memory	K	3
9	Whichoperatorisusedtoaccesstheaddressofavariable? A) * B) & C) % D)@	U	3
10	Whichkeywordisusedto defineastructurein C A) struct B)structureC)recordD)define	K	3

1*10=10

Part B

Short Answer Questions

Answer any 4 QuestionsEachquestioncarries5marks

11	Explain the syntax of if-else statement. Write a program to find greatest among two numbers	U	1
12	DiscussanythreeloopstatementsinCwithsyntaxandexample.	K	1
13	Explain the syntax of one-dimensional array. Write a C program to calculate the sum of all elements in an integer array.	U	2
14	DiscussvariousstringhandlingfunctionsinC	K	2
15	WriteaCprogramusingstructuretostoreanddisplayinformationofastudent. Include fields like name, roll number and mark	A	3
16	ExplainpointersinC.Writeaprogramtoprinttheaddressofanintegervariable using pointer.	U	3

4 * 5 =20

Part C

Essay

QuestionsAnswerany**2**Q uestions

Eachquestioncarries10marks

17	ExplainwithexamplesdifferentoperatorsinC.	K	1
18	Writeaprogramtocountthenumberofvowelsinastring.Alsodrawthe flowchart.	A	2
19	Differentiatebetweenbuiltinfunctionsanduserdefinedfunctions.Whatare recursive functions?	U	2
20	Explainwithsupportingexamplescallbyvalueandcallbyreferencefunction call.	U	3

2 * 10 =20